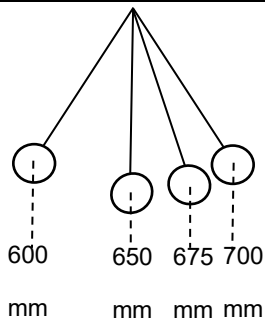


15

B



Taking the initial position at 650 mm mark,

$$x = x_0 \sin \omega t = x_0 \sin \left[\left(\frac{2\pi}{T} \right) t \right] \quad 25 = 50 \sin \left[\left(\frac{2\pi}{2} \right) t \right] \quad t = 0.167 \text{ s}$$

$$x = x_0 \sin \omega t = x_0 \sin \left[\left(\frac{2\pi}{T} \right) t \right] \quad 10 = 50 \sin \left[\left(\frac{2\pi}{2} \right) t \right] \quad t = 0.064 \text{ s}$$

$$t_{\text{Total}} = 0.167 + 0.064 = 0.231 = 0.23 \text{ s}$$

16

D